PROJECT INFORMATION:

Project Name: **Electrical Distribution Resilience Circuit 2 Loop Feed**

Project No: **XTLF 25-1025**

Project Location: VANCE AFB, ENID OKLAHOMA

Project Status: Pre Design

Issue Date: 04 Apr 2025

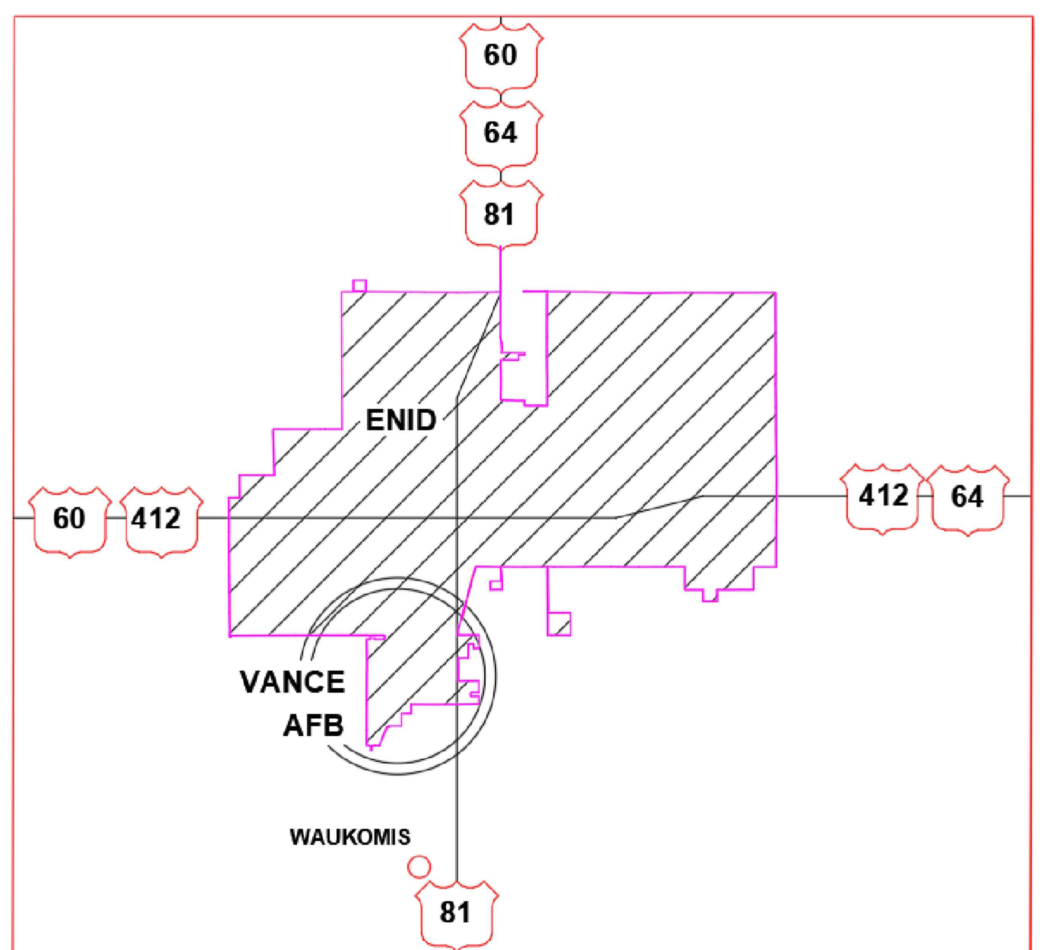
G001

SHEET INDEX

SHEET NUMBER	SHEET NAME	REVISIONS	DATE
G001	TITLE, INDEX & COORDINATION		
G002	PROJECT LOCATION & HAUL ROUTE		
G003	GENERAL NOTES AND LEGENDS		
E101	CIRCUIT 2 LOOP		
E301	ONE LINE DIAGRAM		
E501	SWITCH DETAILS		
E502	MH-5 DETAILS		

RELEVANT CODE DOCUMENTS

DOCUMENT TITLE	
UFC 3-501-01 ELECTRICAL ENGINEERING	2024
UFC 3-550-01 EXTERIOR ELECTRICAL POWER DISTRIBUTION	2016
NFPA 70 - NATIONAL ELECTRIC CODE	2017



VICINITY MAP

NOT TO SCALE

UNITED STATES AIR FORCE AIR EDUCATION TRAINING COMMAND
VANCE AFB, OKLAHOMA

LICENSED PROFESSIONAL ENGINEER
RYAN SUDLOW
31220
OKLAHOMA

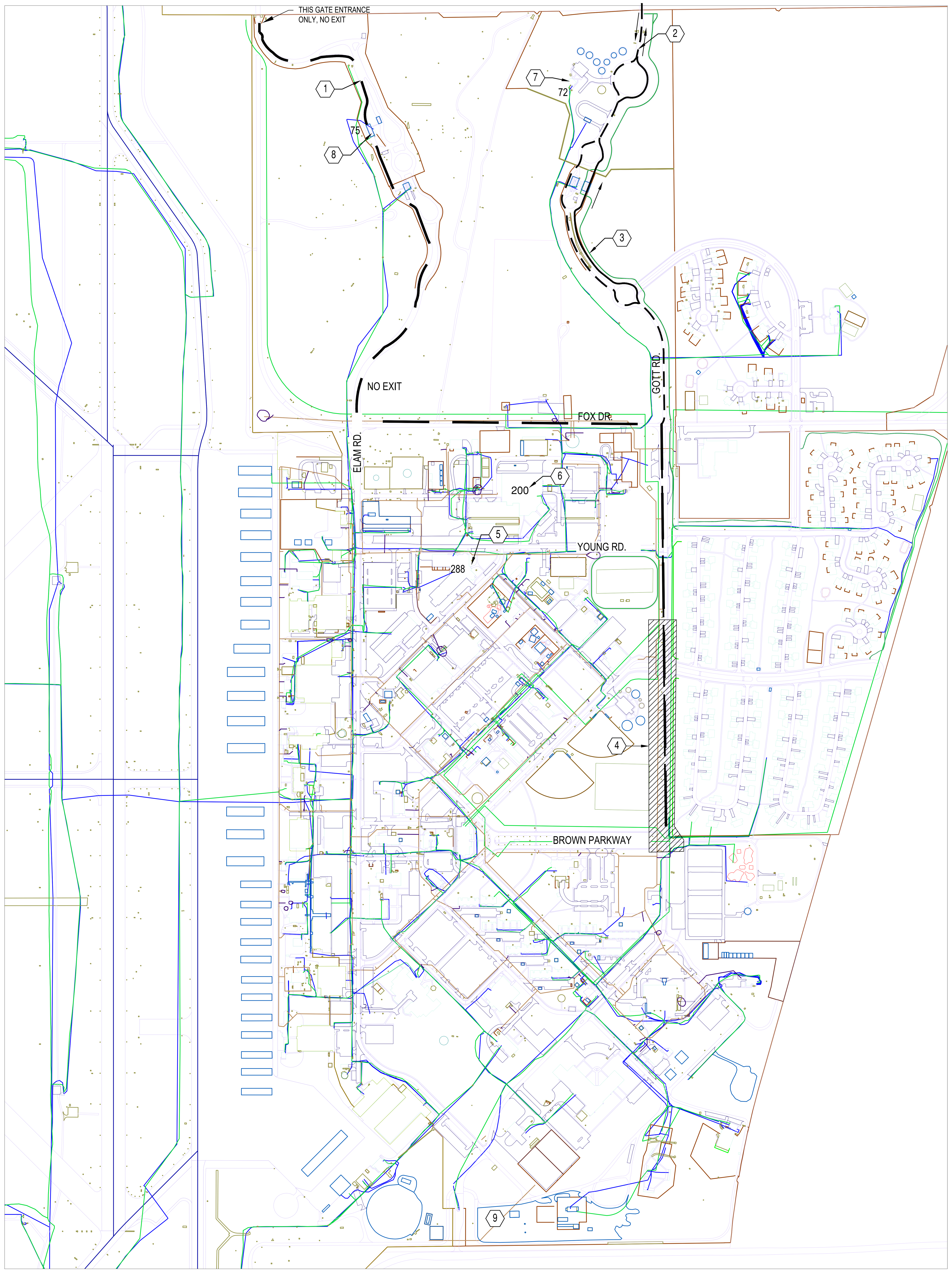
Field Services

140 Channel Street, Suite 205
Vance AFB, Oklahoma 73705
(580) 213-7787
Certificate of Authorization# 8716
Expires: June 30, 2026

EXECUTION OF WORK DESCRIBED WITHIN THIS PLAN SET SHALL BE IAW 48 CFR SUBPART 25.2 - BUY AMERICAN - CONSTRUCTION MATERIALS AND ALL SUBSEQUENT PARTS RELATED TO SUCH REFERENCE.

1 1/2"(ACTUAL SIZE)

DRAWING SIZE INTENDED TO BE 24" x 36"



A
G002
LOCATION AND HAUL ROUTE
NTS FOR LOCATION GENERAL REFERENCE ONLY

N

GENERAL NOTES

1. APPROVED HAUL AND ACCESS ROUTES ARE NOTED ON THE PLAN. THE CONTRACTOR SHALL USE ONLY ACCESS ROUTES THAT ARE PREVIOUSLY APPROVED BY CONSTRUCTION MANAGEMENT.
2. CONSTRUCTION MANAGERS OFFICE IS LOCATED ON THE FIRST FLOOR NORTH-WEST CORNER OF BUILDING 288.
3. CONTRACTING IS LOCATED IN BUILDING 500.
4. REFER TO SPECIFICATION SECTION 010000 - GENERAL PROVISIONS FOR BASE ACCESS PROCEDURES AND RULES.

LEGEND

PROJECT LOCATION

PRIMARY INDUSTRIAL INSPECTION ENTRY

SECONDARY BASE ENTRY AND ALL VEHICLE EXIT ROUTE

KEYNOTES

1. PRIMARY HAUL ROUTE.
2. SECONDARY HAUL ROUTE TO BE USED ONLY WHEN PRIMARY ROUTE GATE IS CLOSED.
3. MAIN GATE ENTRANCE ONLY AND ALL VEHICLE EXIT
4. PROJECT LOCATION.
5. FACILITY 288 CIVIL ENGINEERING INCLUDING CONSTRUCTION MANAGEMENT.
6. FACILITY 200 PROCUREMENT.
7. FACILITY 72 VISITOR'S CENTER, USE FOR PASS APPROVAL PRIOR TO BEGINNING WORK.
8. FACILITY 75 COMMERCIAL VEHICLE INSPECTION.
9. LAYDOWN AREA, COORDINATE WITH CONSTRUCTION MANAGER.

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(580) 213-7772
CA# 8716 EXPIRES
JUNE 30, 2026

Revision Number	Description	Date

LICENSED PROFESSIONAL ENGINEER
RYAN SUDLOW
31220
OKLAHOMA
17 APR 2025

DRAWN BY: RLS
CHECKED BY: RLS
PROJECT NO.: XTLF 25-1025
DATE: 04 APR 2025
DESIGN STAGE: FOR CONSTRUCTION

ELECTRICAL DISTRIBUTION RESILIENCE
CIRCUIT 2 LOOP FEED

PROJECT LOCATION & HAUL ROUTE

G002

OVERALL PROJECT GENERAL NOTES

1. SECTIONS OF THE NEW DISTRIBUTION SYSTEM ARE TO BE IN PLACE AND FULLY TESTED ACCORDING TO THE PROJECT SPECIFICATIONS PRIOR TO DEMOLITION OF THE EXISTING DISTRIBUTION SYSTEM TO BE REPLACED. REFERENCE THE PROJECT SCHEDULE F.
2. PROVIDE A 21 DAY NOTICE INCLUDING A LIST OF AFFECTED FACILITIES AND AREAS TO VANCE AFB CONSTRUCTION MANAGEMENT PRIOR TO ELECTRICAL SERVICE DISCONNECTS OR DISRUPTIONS.. .
3. EXISTING UTILITIES WERE TAKEN INTO CONSIDERATION DURING THE DESIGN PROCESS HOWEVER FOR CLARITY ONLY EXISTING AND PROPOSED ELECTRICAL LINES ARE SHOWN ON THE FOLLOWING PLAN SHEETS. IT SHALL BE THE RESPONSIBILITY OF CONTRACTOR TO CONTACT APPROPRIATE AGENCIES TO LOCATE ALL UNDERGROUND UTILITIES AND LOCATIONS PRIOR TO DIGGING.
4. DURING THE DURATION OF THIS PROJECT IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO MAINTAIN MARKINGS AND FLAGS FOR EXISTING UTILITY LOCATIONS AND TO NOTIFY CONSTRUCTION MANAGEMENT PRIOR TO ANY CUTTING OR DISTURBANCE OF EXISTING UTILITIES.
5. UNLESS INDICATED ON PLANS, ALL OPEN TRENCHING OF ROADWAYS AND PARKING AREAS MUST BE APPROVED BY BASE CIVIL ENGINEERING. OPEN TRENCHING OR ROADWAY CUTS WILL NOT BE PERMITTED FOR CROSSING OF ELAM RD.
6. FOR ALL APPROVED ROADWAY, PARKING, AND SIDEWALK CUTS REFERENCE DETAIL SHEETS C501 AND C502 FOR MATERIAL AND REPAIR METHOD REQUIREMENTS. WHERE MULTIPLE ENTRANCES EXIST FOR PARKING LOTS, ONLY ONE MAY BE CUT PER PARKING LOT AT ANY ONE TIME TO ALLOW CONTINUED USE OF THAT PARKING LOT.
7. THE CONTRACTOR SHALL RESTORE ALL DISTURBED GRAVEL AREAS, REFERENCE DETAIL H/C501.
8. THE CONTRACTOR SHALL RESTORE SOD IN ALL DISTURBED TURF AREAS. IF MORE THAN ONE ACRE OF LAND IS TO BE DISTURBED, CONTRACTOR MUST OBTAIN STORMWATER PERMIT.
9. ALL WASTE PRODUCTS EXCEPT HAZARDOUS WASTE SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND DISPOSED OFF SITE IN ACCORDANCE WITH THE SPECIFICATIONS. HAZARDOUS WASTE DISPOSAL SHALL BE COORDINATED THROUGH THE CONSTRUCTION MANAGER.
10. ALL EQUIPMENT IS TO BE PROVIDED NEW UNLESS NOTED OTHERWISE.
11. THE LOCATION OF PROPOSED SWITCHGEAR, SECTIONALIZERS, AND CAPACITORS MAY BE ADJUSTED TO AVOID CONFLICTS AS REQUIRED, FOLLOWING APPROVAL BY THE ENGINEER AND CONSTRUCTION MANAGER.
12. INTERCEPTION LOCATIONS OF EXISTING UNDERGROUND PRIMARY AND SECONDARY MAY BE ADJUSTED TO AVOID CONFLICTS AS REQUIRED, FOLLOWING APPROVAL BY THE ENGINEER AND CONSTRUCTION INSPECTION MANAGER.
13. IF ANY TREES ARE REMOVED TO ALLOW FOR CONSTRUCTION OF THIS PROJECT, A NEW TREE MUST BE PLANTED. COORDINATE WITH CONSTRUCTION MANAGER.
14. BASIS OF DESIGN FOR THE EQUIPMENT IN THIS PROJECT IS AS FOLLOWS:
    - 14.1. EATON'S COOPER POWER: VFI UNDERGROUND DISTRIBUTION SWITCHGEAR, SINGLE AND THREE PHASE SECTER SECTIONALIZING CABINETS, SINGLE AND THREE PHASE PAD MOUNTED COMPARTMENTAL TYPE TRANSFORMERS, METAL ENCLOSED, PAD MOUNTED CAPACITOR BANKS.
    - 14.2. SQUARE D: DISTRIBUTION PANELS, BREAKER TYPE DISCONNECTS
    - 14.3. STRESSCRETE: SPUN CONCRETE LIGHT POLES
    - 14.4. VISIONAIRE: BOW SX LED LUMINAIRES
15. SHEET NUMBERING ADHERES TO THE FOLLOWING CONVENTION:
    - 15.1. E101X / E102X
      - 15.1.1. X DENOTES SYSTEM SHOWN AS FOLLOWS:
        - 15.1.1.1. P – PRIMARY MEDIUM VOLTAGE DISTRIBUTION
        - 15.1.1.2. S – SECONDARY LOW VOLTAGE DISTRIBUTION
        - 15.1.1.3. D – DEMOLITION OF EXISTING INFRASTRUCTURE

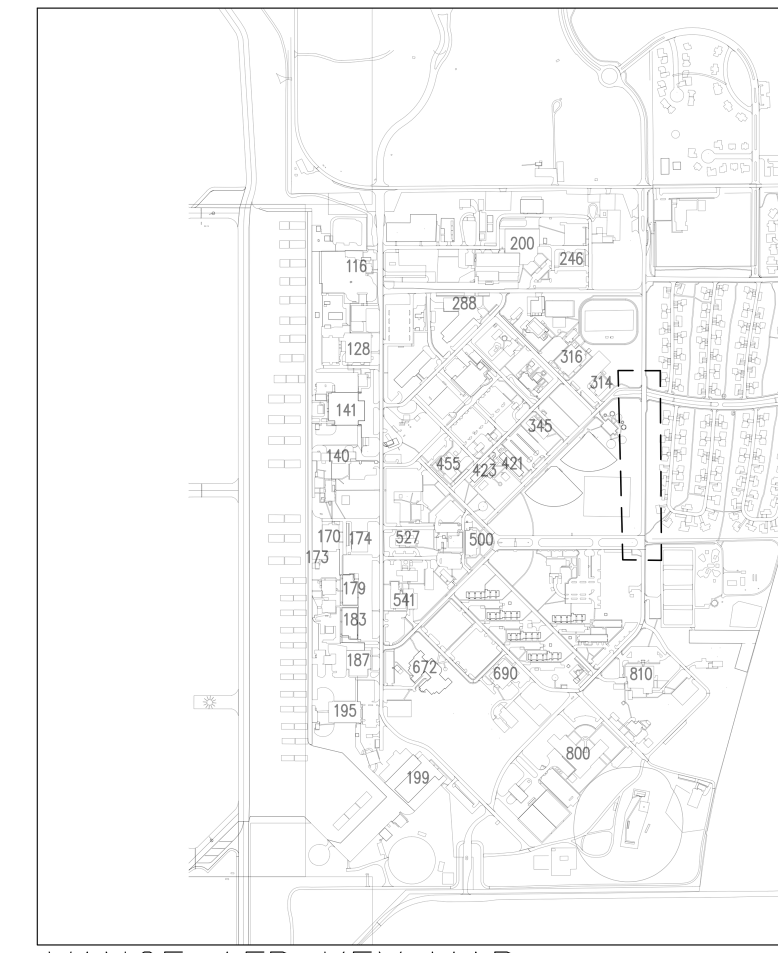
LEGEND

NEW PRIMARY

EU	ELEC. PRIMARY UNDERGROUND (U/G) OPEN TRENCH (RE: C/E101)
	PAD MOUNTED TRANSFORMER (T)
SW	PAD MOUNTED SWITCH (SW)
S	SECTIONALIZER (PAD MOUNTED JUNCTION BOX)
	DIRECTIONAL BORE
	CUT & REPLACE CONCRETE
	POWER POLE
	PAD MOUNT CAPACITOR

EXISTING

EUx EUx	ELECTRICAL PRIMARY U/G	Eh	ELECTRIC HANDHOLE
Ex	ELECTRICAL (O/H)	Ej	ELECTRIC JUNCTION BOX
ESx ESx	ELECTRICAL SECONDARY (U/G)		ELECTRIC SUBSTATION
ES ES	ELECTRICAL SECONDARY (O/H)	SW	ELECTRIC SWITCH
	IRP AREA		ELECTRIC CAPACITOR
X Gx X	CHAINLINK OR METAL FENCE		ELECTRIC SECTIONALIZER
	CONCRETE WALL / FENCE	mt	ELECTRIC METER
	EXISTING FACILITY		ELECTRIC POLE
	PAVED SURFACE	G	GENERATOR
	PAD MOUNTED TRANSFORMER	GV	GIANT VOICE TOWER
	POLE MOUNTED TRANSFORMER	cp	CATHODIC PROTECTION RECTIFIER
	LIGHT POLE		TREE
Em	ELECTRIC MANHOLE		EXISTING ELECTRIC POLE TO BE REMOVED, RE: DEMO SHEET



VANCE AFB KEY MAP
NOT TO SCALE

A
G003
EXISTING UTILITIES
1" = 100'

N

0 100' 200'
SCALE: 1" = 100'

CONCRETE GENERAL NOTES

1. DESIGN IS IN ACCORDANCE WITH THE FOLLOWING DOCUMENTS:
   - UFC 1–200–01 DOD BUILDING GUIDE
   - UFC 1–200–02 HIGH PERFORMANCE AND SUSTAINABLE BUILDING REQUIREMENTS
   - UFC 3–201–01 CIVIL ENGINEERING
   - UFC 3–301–01–2013–C4 STRUCTURAL ENGINEERING.
   - UFC 3–220–01 GEOTECHNICAL ENGINEERING
   - UFC 3–260–01 SAFETY AND SECURITY
   - UFC 4–010–01 DOD MINIMUM ANTITERRORISM STANDARDS FOR BUILDINGS
   - UFC 4–020–01–2008 DOD SECURITY ENGINEERING FACILITIES PLANNING MANUAL.
   - AMERICAN SOCIETY OF CIVIL ENGINEERS, ASCE 7–16, PROVISIONS AND COMMENTARY.
   - AMERICAN CONCRETE INSTITUTE, ACI 318–08.
2. VERIFY ALL DIMENSIONS, ELEVATIONS, AND SITE CONDITIONS PRIOR TO STARTING CONSTRUCTION. NOTIFY THE ENGINEER IF ANY DISCREPANCIES OR INCONSISTENCIES ARE FOUND.
3. SEE STRUCTURAL, MECHANICAL AND ELECTRICAL DRAWINGS FOR INFORMATION NOT SHOWN IN THIS SECTION.
4. SUBMITTALS SHALL CONFORM TO REQUIREMENTS OF CONTRACT DOCUMENTS AND SHALL BE CHECKED BY THE CONSTRUCTION CONTRACTOR'S QUALITY CONTROL REPRESENTATIVE PERSON TO SUBMISSION TO CONSTRUCTION MANAGER.
5. COORDINATE ALL DIMENSIONS WITH OTHER SECTIONS OF THIS PLAN SET. NOTIFY CONSTRUCTION MANAGER OF ANY CONFLICTS PRIOR TO CONSTRUCTION.
6. PROJECT SPECIFICATIONS ARE PART OF THE CONSTRUCTION DOCUMENT AND ARE TO BE USED IN CONJUNCTION WITH THE DRAWINGS.
7. VERIFY ALL CONDITIONS, EXISTING AND NEW, SHOWN ON THE CONSTRUCTION DOCUMENTS PRIOR TO PROCEEDING WITH WORK. DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE CONSTRUCTION MANAGER. AFFS IS NOT RESPONSIBLE FOR WORK DONE IN THESE AREAS WITHOUT CLARIFICATION IN WRITING FROM THE CONSTRUCTION MANAGER.
8. ALL PHASES OF CONSTRUCTION SHALL CONFORM TO THE MINIMUM STANDARDS OF THE BUILDING CODES NOTED ON THIS SHEET.
9. DIMENSIONS SHOWN ON CONSTRUCTION DOCUMENTS TAKE PRIORITY OVER SCALED DIMENSIONS. IN SOME CASES, PLANS AND DETAILS MAY NOT BE DRAWN TO SCALE FOR CLARITY.
10. DO NOT LOAD THE CONCRETE SLAB–ON–GRADE WITH EQUIPMENT UNTIL SUFFICIENT CONCRETE STRENGTH HAS BEEN OBTAINED AND VERIFIED BY TESTING.
11. THIS PLAN SECTION TO BE USED IN CONJUNCTION WITH THE OTHER SECTION FOR THIS PROJECT. CONTRACTOR RESPONSIBLE FOR IMPLEMENTING THE INFORMATION SHOWN ON ALL REFERENCED PLANS.
12. CONTRACTOR IS RESPONSIBLE FOR MEANS AND METHODS OF DEMOLITION AND CONSTRUCTION. THE CONSTRUCTION SEQUENCE SHOULD NOT IMPACT THE FINAL DESIGN SHOWN ON THESE DOCUMENTS AND APPROVED SUBMITTALS.

GENERAL CAST–IN–PLACE CONCRETE NOTES

1. ALL CONCRETE WORK SHALL CONFORM TO THE CURRENT ACI 318 BUILDING CODE AND COMMENTARY. REFER TO PROJECT SPECIFICATIONS SECTION 03 30 00 CAST–IN–PLACE CONCRETE FOR ADDITIONAL INFORMATION.
2. CONCRETE SLAB SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 4,000 PSI AT 28–DAYS.
3. FORMWORK TOLERANCES SHALL BE IN ACCORDANCE WITH ACI 347 SPECIFICATIONS.
4. CONCRETE SHALL BE PROPORTIONED SUCH THAT THE WATER–CEMENT RATIO (BY WEIGHT) DOES NOT EXCEED 0.5. CONCRETE MUST CONTAIN A MINIMUM OF SIX SACKS OF CEMENT PER CUBIC YARD OF CONCRETE.
5. CEMENT MUST BE PORTLAND CEMENT IN ACCORDANCE WITH ASTM C150, TYPE I/II OR TYPE III.
6. AGGREGATES SHALL BE GRADED IN ACCORDANCE WITH ASTM C33 SPECIFICATIONS. COURSE AGGREGATE SHALL BE NO GREATER THAN 3/4" NOMINAL DIAMETER. FINE AGGREGATE SHALL BE NATURAL SAND FREE FROM MATERIALS WITH DELETERIOUS REACTIVITY TO ALKALI IN CEMENT.
7. WATER SHALL BE IN ACCORDANCE WITH ASTM C94 AND POTABLE.
8. AIR CONTENT SHALL BE BETWEEN 5% AND 7% (BY VOLUME).
9. ADMIXTURES ARE ALLOWED.
10. CURING SHALL BE ACCOMPLISHED BY WET CURING OR APPLICATION OF TYPE 2 MEMBRANE.
11. REINFORCING STEEL SHALL BE DEFORMED, NEW BILLET BARS PER ASTM A615 SPECIFICATIONS AND MEET GRADE 60 REQUIREMENTS.
12. FABRICATION OF REINFORCING STEEL SHALL BE CHAPTER 7 OF THE CONCRETE REINFORCING STANDARDS INSTITUTE MANUAL FOR STANDARD PRACTICE.
13. REINFORCING STEEL SHALL BE BLOCKED AND TIED TO PROPER LOCATION AND SECURELY WIRED AGAINST DISPLACEMENT. TIE WIRES SHALL BE INSTALLED AT EVERY OTHER BAR INTERSECTION SO THAT AT LEAST 50% OF THE INTERSECTIONS ARE TIED IN THE FOOTINGS AND SLABS. TACK WELDING OF REINFORCING IS PROHIBITED.
14. ALL REINFORCING SHALL BE CONTINUOUS. LAP REINFORCING BARS FOR A MINIMUM LENGTH OF 40 TIMES THE BAR DIAMETER. ALTERNATE SPLICE LOCATIONS WHERE POSSIBLE. LAPS IN VERTICAL BARS IN WALL ARE ALLOWED AT THE BOTTOM AND AT THE MID WALL HEIGHT. ALTERNATE SPLICE LOCATIONS.
15. MINIMUM CONCRETE REINFORCING SHALL BE AS FOLLOWS:
    - a. 3" CLEAR WHEN CONCRETE IS AGAINST EARTH OR GRADE.
    - b. 2" AT ALL OTHER LOCATIONS.
16. DO NOT WORK ON NEW SLAB UNTIL CONCRETE HAS REACHED SUFFICIENT STRENGTH, AS DETERMINED BY CONCRETE STRENGTH TESTS TO SUPPORT NEW EQUIPMENT.

SLAB PREPARATION

1. PREPARE SOIL IN ACCORDANCE WITH PLANS AND PROJECT SPECIFICATIONS SECTION 31 00 00 EARTH MOVING.
2. DESIGN FROST DEPTH: 22 INCHES (PER UFC 3–301–01 TABLE E–2).
3. REMOVE FREE WATER FROM EXCAVATIONS AND DRY OUT BEFORE PLACING CONCRETE.
4. DESIGN BEARING PRESSURE IS 2000 PSF FOR SLAB BEARING ON ODOT TYPE "A" AGGREGATE BASE COMPACTED TO 95% ASTM D–1557 DENSITY. ALLOWABLE BEARING PRESSURE TO BE DETERMINED BY PROFESSIONAL GEOTECHNICAL ENGINEER.
5. GEOTEXTILE SEPARATION FABRIC SHALL HAVE PROPERTIES EQUIVALENT TO HEAVY DUTY WOVEN FABRICS SUCH AS MIRAFI RS600 OR TERRATEX HPG–57.
6. FLOOR SLAB REINFORCING SHALL BE SUPPORTED AT 24"ON CENTER SPACING BY FLAT BOTTOMED CHAIRS OR CONCRETE BRIQUETTES OF 4000 PSI CONCRETE.

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CA# 8716 EXPIRES
JUNE 30, 2026

Revision Number	Description	Date

LICENSED PROFESSIONAL ENGINEER
RYAN SUDLOW
31220
OKLAHOMA
17APR2025

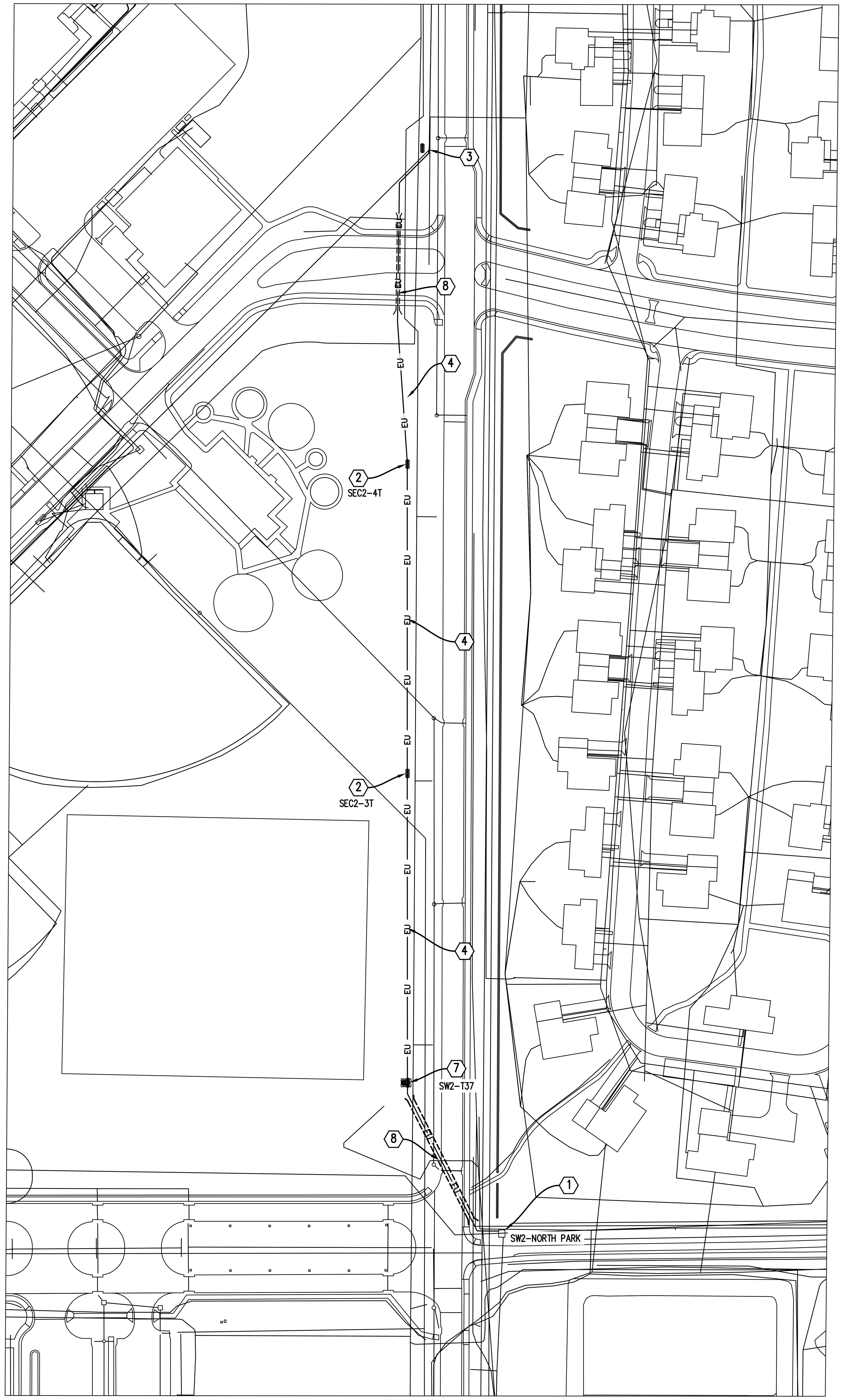
DRAWN BY: RLS
CHECKED BY: RLS
PROJECT NO.: XTLF 25-1025
DATE: 04 APR 2025
DESIGN STAGE: FOR CONSTRUCTION

ELECTRICAL DISTRIBUTION RESILIENCE
CIRCUIT 2 LOOP FEED

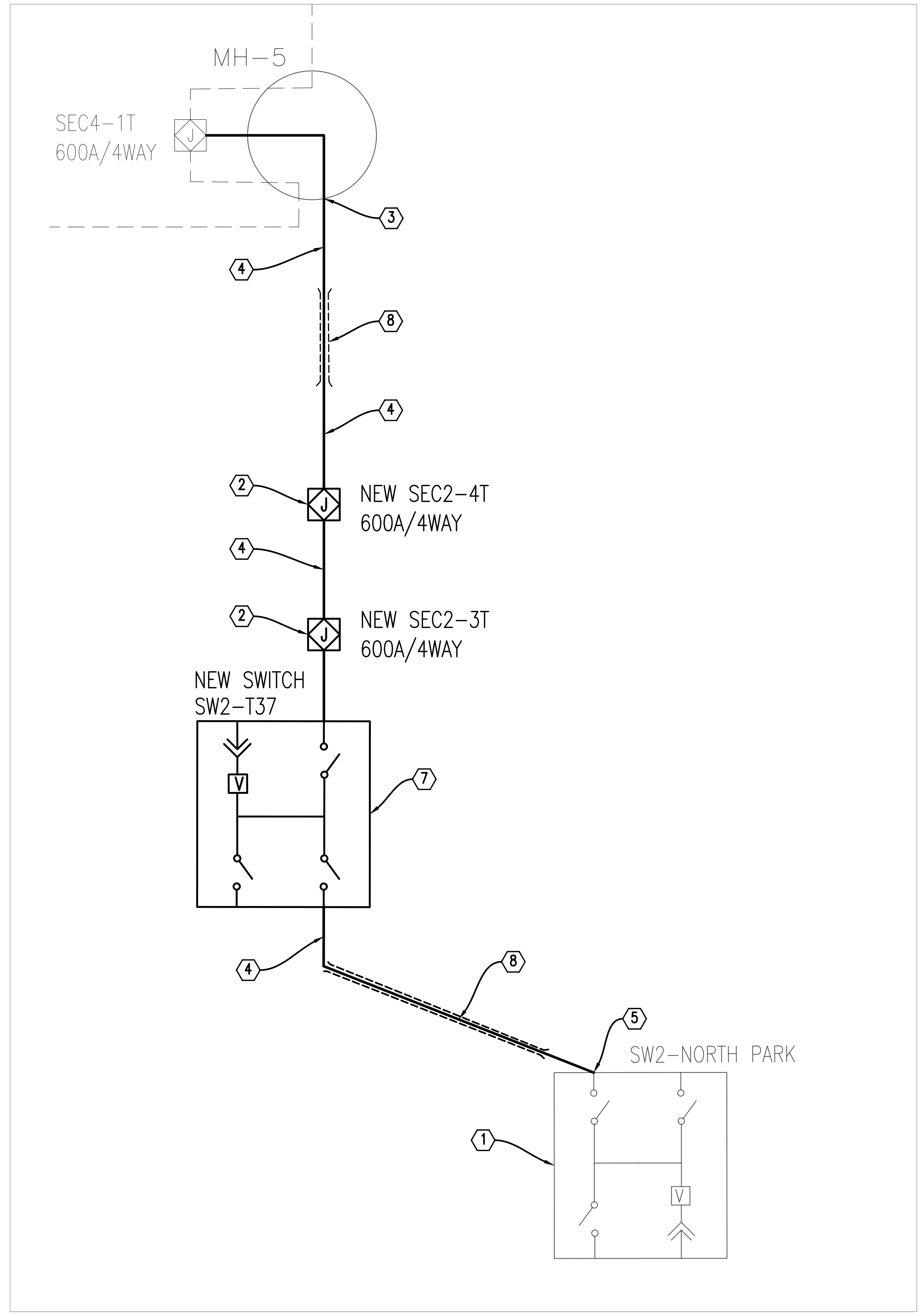
3

GENERAL NOTES AND LEGENDS

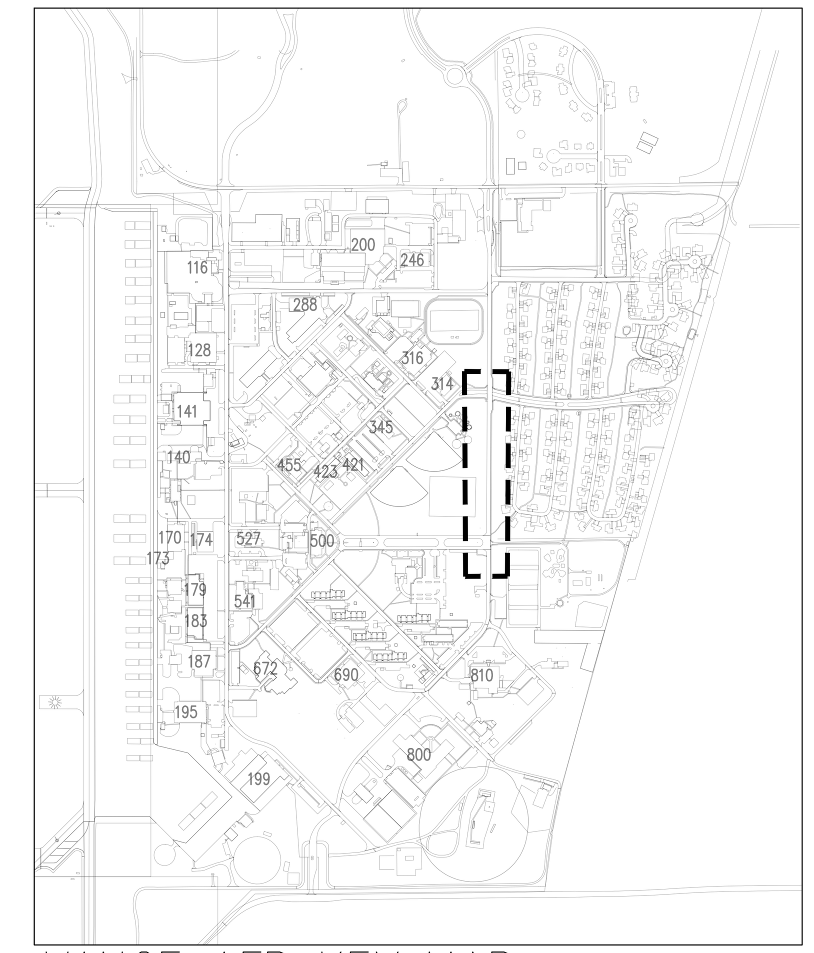
G003



A E101 CIRCUIT 2 LOOP FEED LAYOUT
1" = 75'



B E101 CIRCUIT 2 LOOP FEED ONE-LINE DIAGRAM
NOT TO SCALE



VANCE AFB KEY MAP
NOT TO SCALE

GENERAL NOTES

1. ALL WIRE SHOWN ON THIS SHEET BETWEEN SWITCHES AND/ OR 600A SECTIONALIZERS SHALL BE 250 MCM CU AND SHALL BE IN 5-IN SCHEDULE 40 PVC CONDUIT FOR OPEN TRENCH INSTALLATION OR 5-IN SDR9 HDPE PIPE FOR HORIZONTAL DIRECTIONAL DRILL INSTALLATION.
2. ALL CONDUIT INSTALLED IN THIS PROJECT SHALL HAVE A MINIMUM OF AT LEAST ONE SPARE EMPTY CONDUIT INSTALLED OF THE SAME SIZE, CONNECTING TO THE SAME EQUIPMENT. PULL STRING SHALL BE INCLUDED IN ALL SPARE DUCTS.

KEYNOTES

1. EXISTING TYPE 11 PAD MOUNTED SWITCHGEAR SW2-NORTH PARK
2. NEW COOPER 15KV 600A 4WAY 3ϕ SECTIONALIZER
3. CONNECT INTO EXISTING MANHOLE AND FEED NEW LINE TO SEC4-1T
4. NEW 2X1 5-IN PVC CONDUIT DUCT BANK
5. CONNECT NEW CONDUIT INTO SWITCH VAULT.
6. CONNECT INTO MH-5 FOR CONNECTION INTO SEC4-1T
7. NEW TYPE 11 COOPER PAD MOUNTED SWITCHGEAR
8. HORIZONTAL DIRECTIONAL DRILL DUCT WITH 5-IN SDR9 HDPE PIPE.

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RYAN SUDLOW
31220
OKLAHOMA
17APR2025

DRAWN BY: RLS CHECKED BY: RLS
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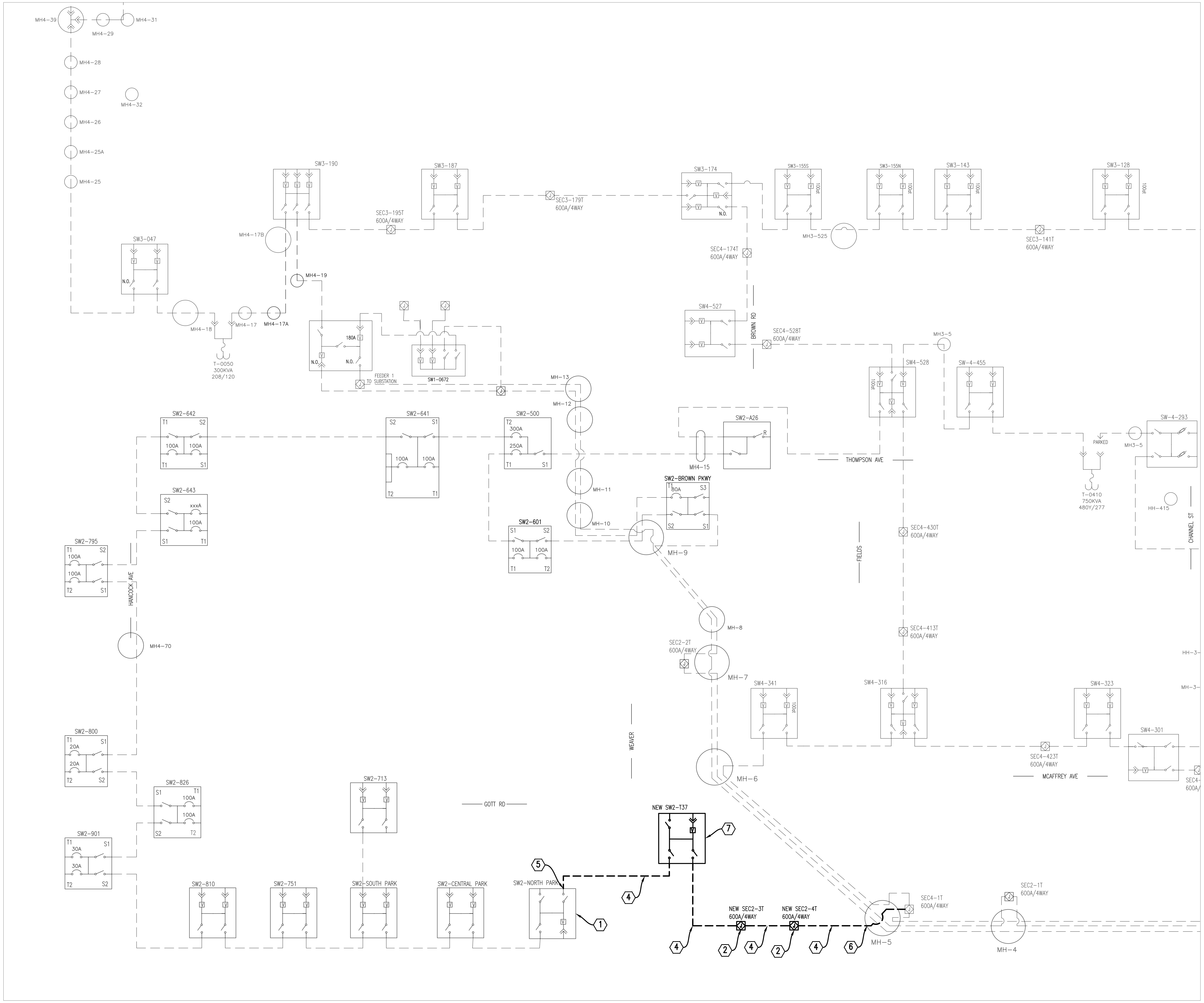
DESIGN STAGE: FOR CONSTRUCTION

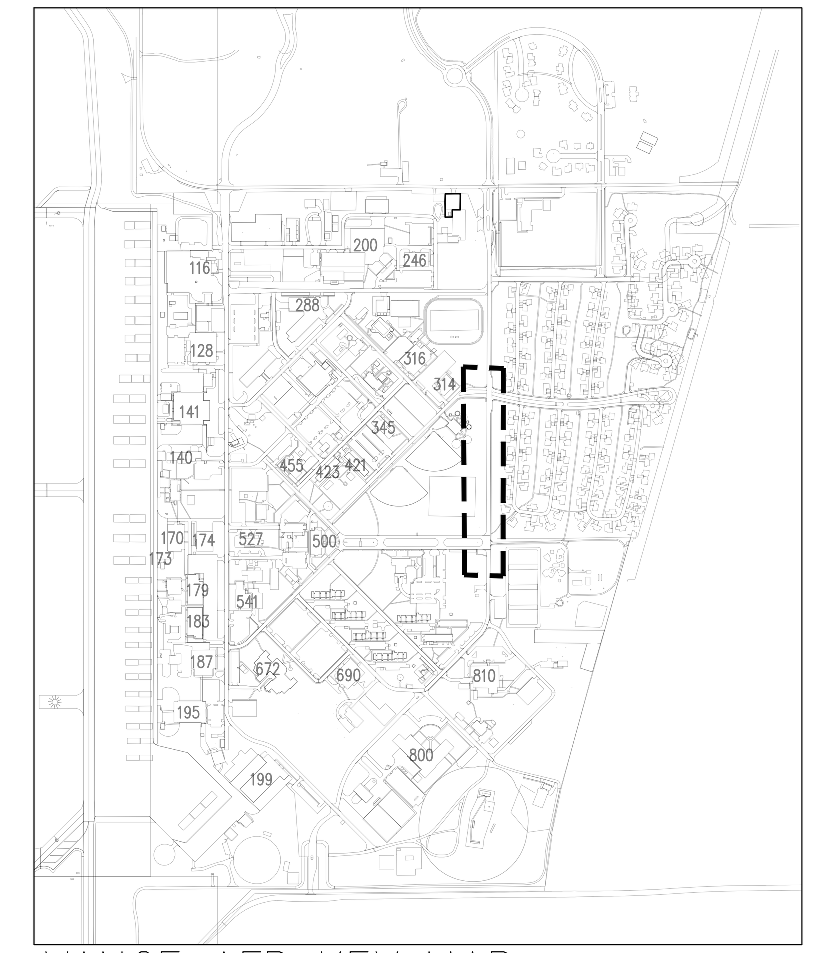
ELECTRICAL DISTRIBUTION RESILIENCE CIRCUIT 2 LOOP FEED

CIRCUIT 2 LOOP FEED

E101



A E301 ONE LINE DIAGRAM
NOT TO SCALE



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LICENSED PROFESSIONAL ENGINEER RYAN SUDLOW 31220 OKLAHOMA 4/7/2025

DRAWN BY: RLS
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PROJECT NO.: XTLF 25-1025
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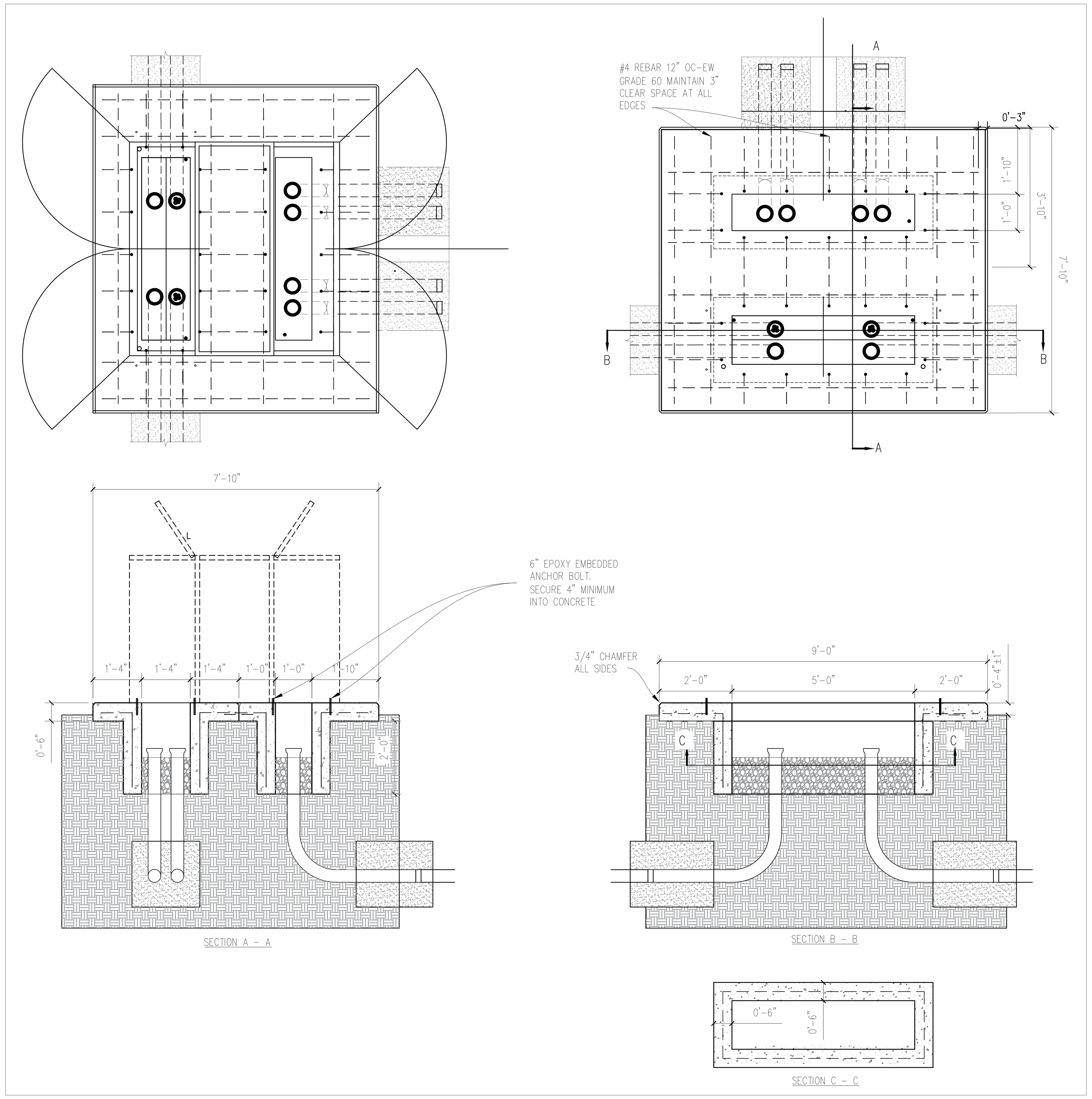
ELECTRICAL DISTRIBUTION RESILIENCE CIRCUIT 2 LOOP FEED

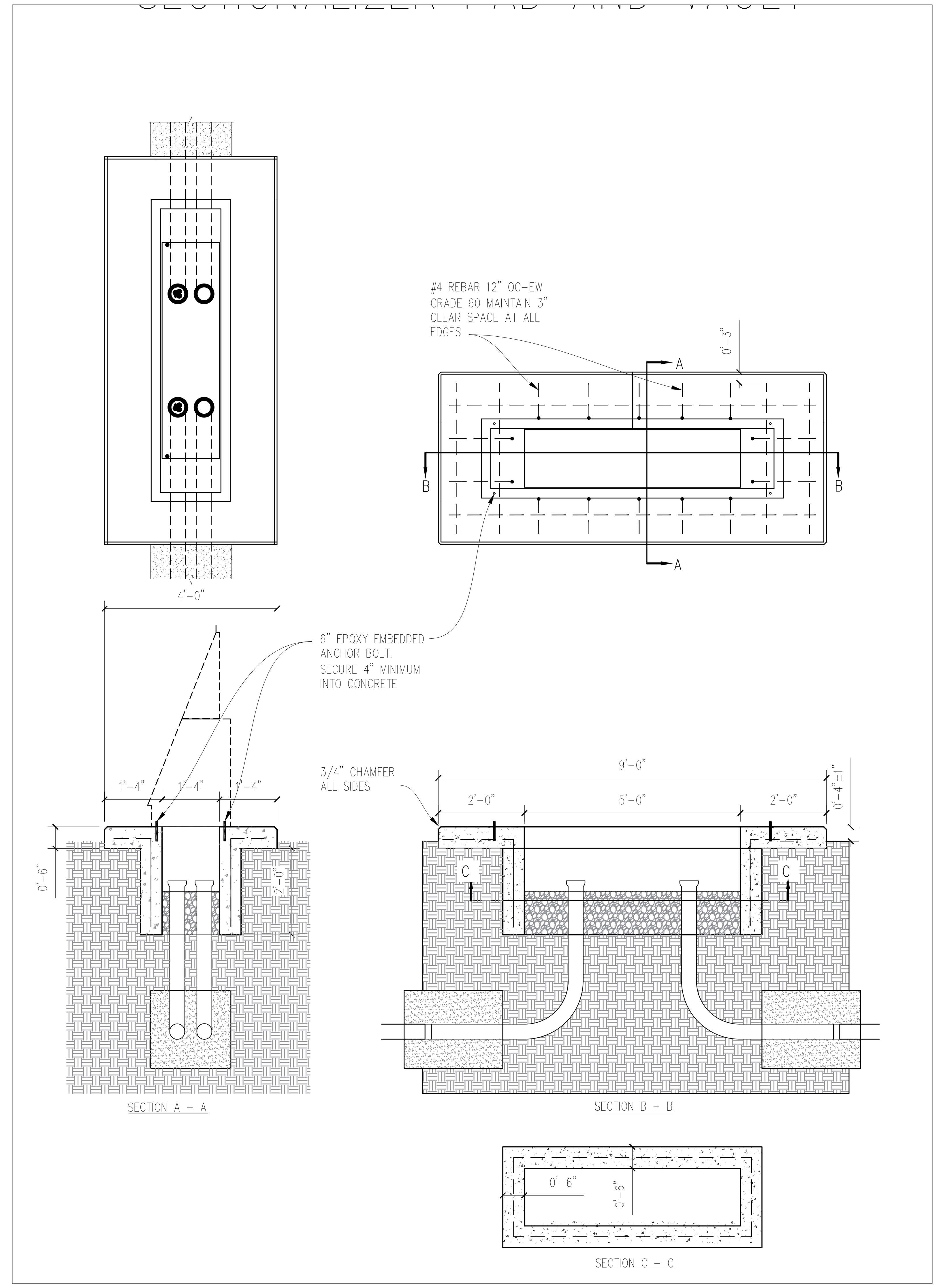
ONE LINE DIAGRAM

E501



A E501 SWITCH PAD DETAILS

NOT TO SCALE



B E501 SECTONALIZER PAD DETAIL

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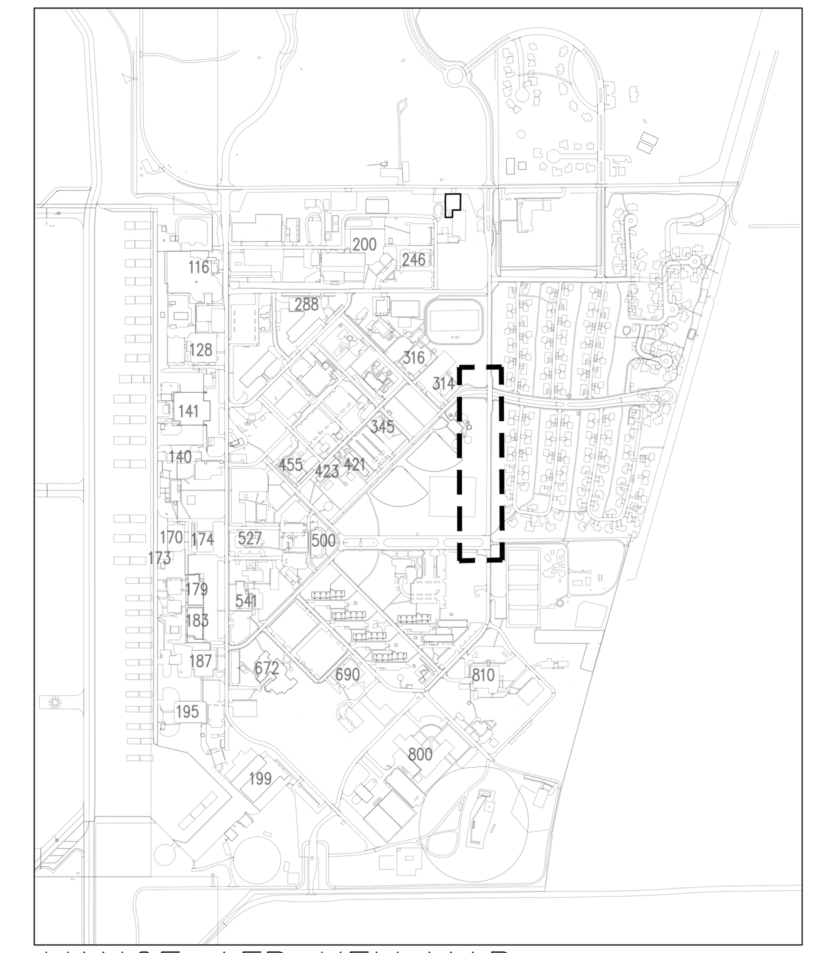
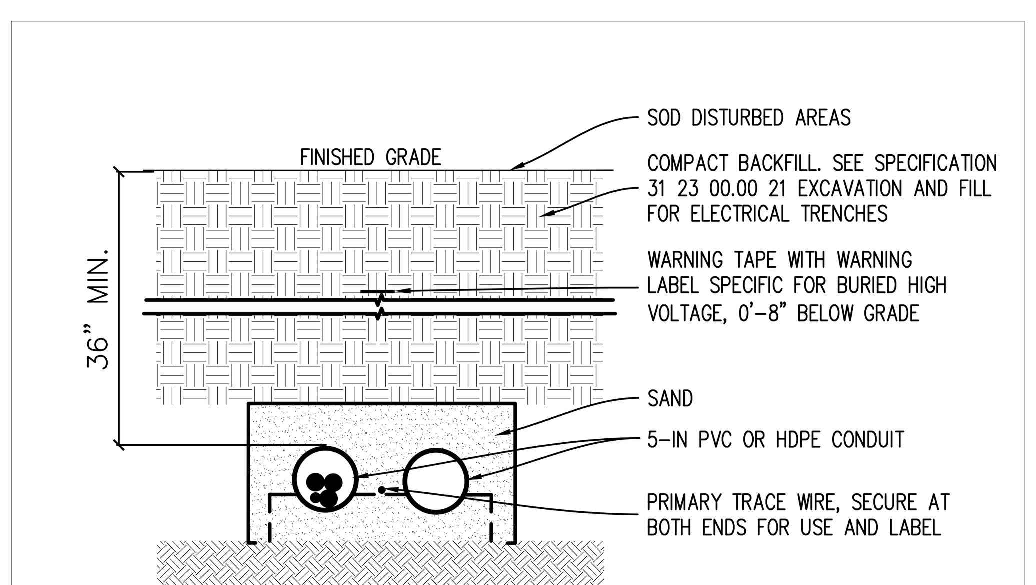


VANCE AFB KEY MAP

N.T.S.

GENERAL NOTES

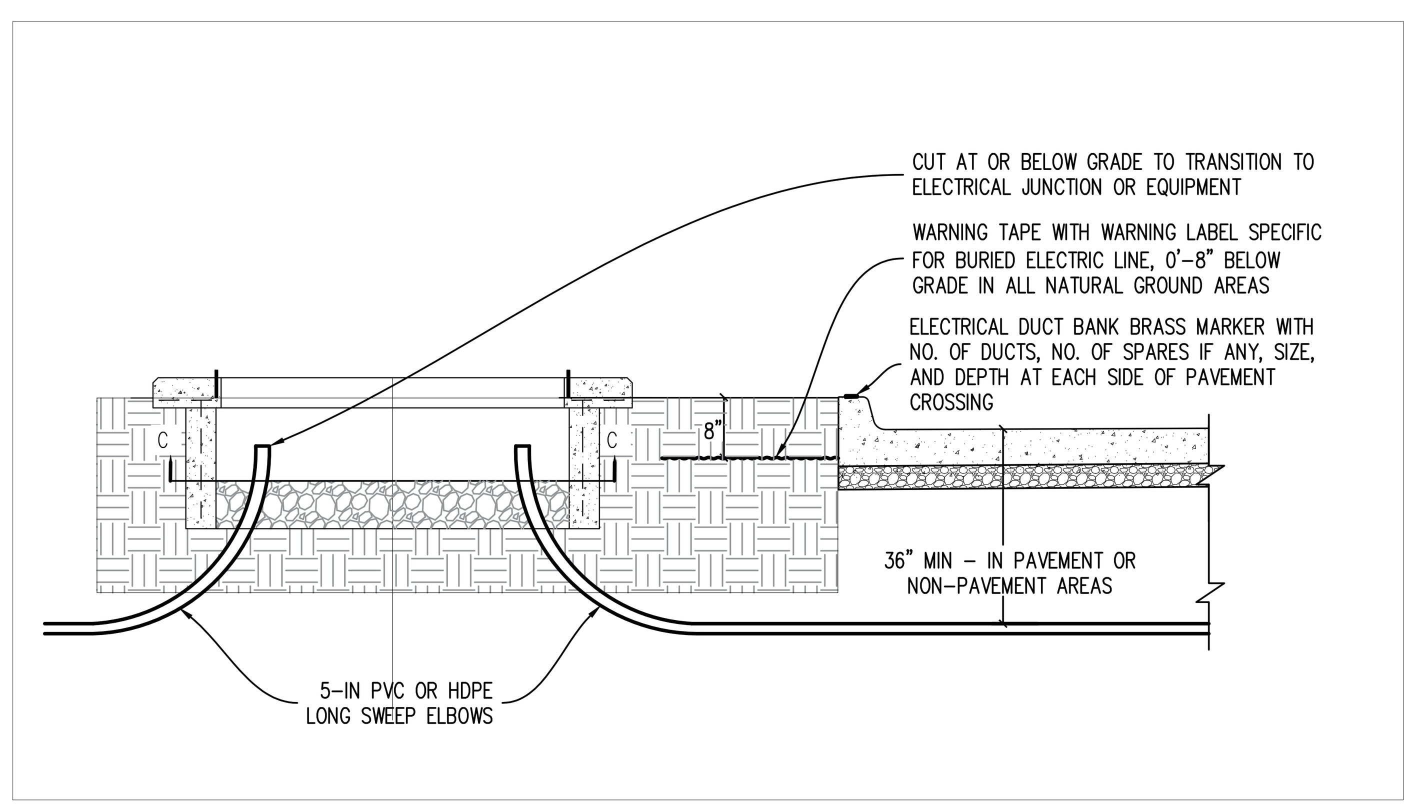
1. ALL WIRE SHOWN ON THIS SHEET BETWEEN SWITCHES AND/ OR 600A SECTIONALIZERS SHALL BE 250 MCM CU AND SHALL BE IN 5-IN SCHEDULE 40 PVC CONDUIT FOR OPEN TRENCH INSTALLATION OR 5-IN SDR9 HDPE PIPE FOR HORIZONTAL DIRECTIONAL DRILL INSTALLATION.
2. ALL CONDUIT INSTALLED IN THIS PROJECT SHALL HAVE A MINIMUM OF AT LEAST ONE SPARE EMPTY CONDUIT INSTALLED OF THE SAME SIZE, CONNECTING TO THE SAME EQUIPMENT. PULL STRING SHALL BE INCLUDED IN ALL SPARE DUCTS.



NOTES:
1. WHEN MORE THAN ONE DUCT/CONDUIT IS INSTALLED, MAINTAIN A 4" MINIMUM SEPARATION BETWEEN CONDUITS.
2. WHERE SOIL IS CONSIDERED HIGHLY CORROSIVE (less than 10,000 OHM-CM RESISTANBCE), THE SIZE OF THE COUNTERPOISE SHALL BE #1/0 AWG.

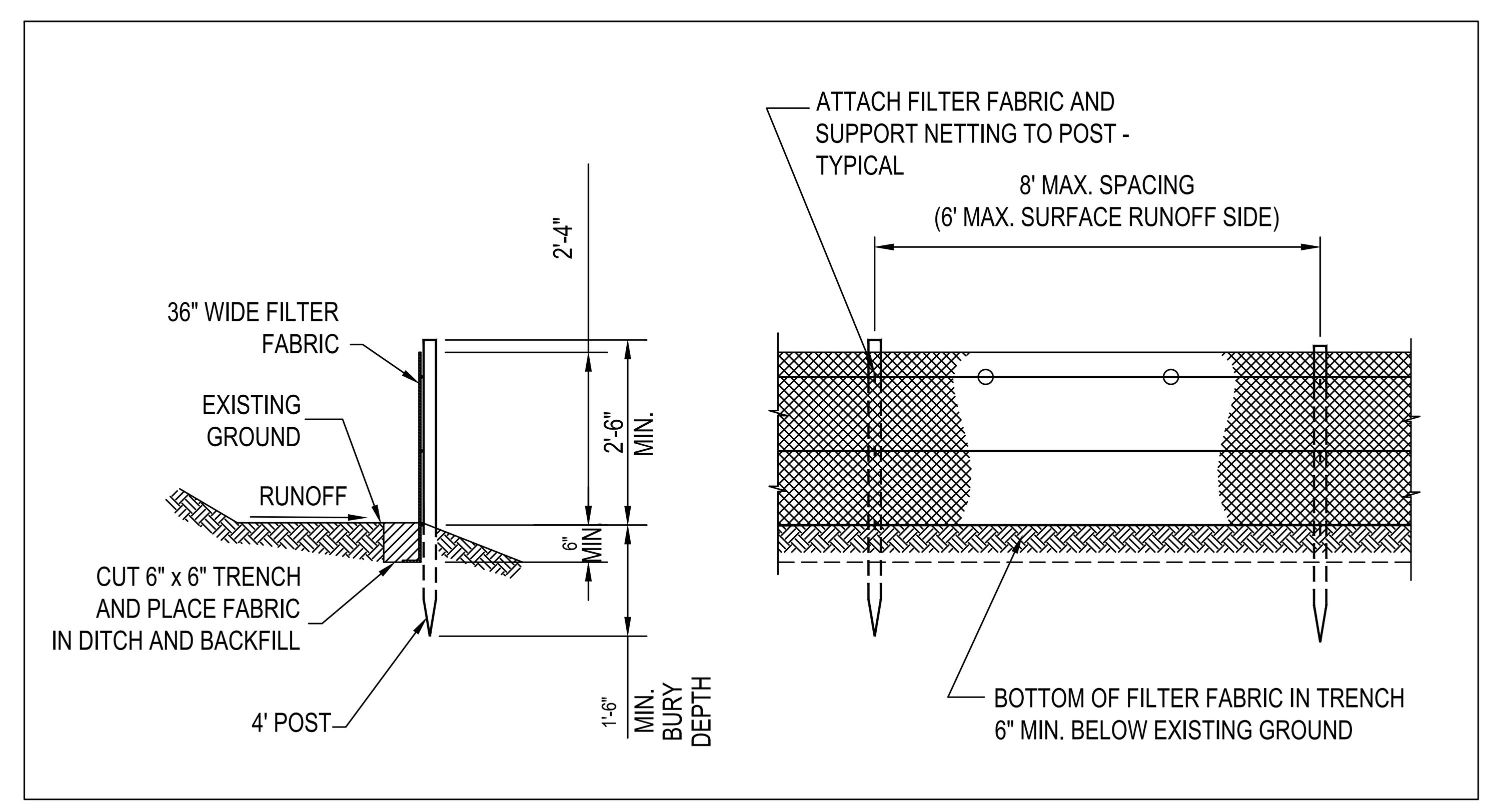
C E501 DUCT BANK DETAIL

NOT TO SCALE



D E501 DUCT BANK TO SECTIONALIZER DETAIL

NOT TO SCALE



E E501 STORM WATER POLLUTION PREVENTION SILT FENCE

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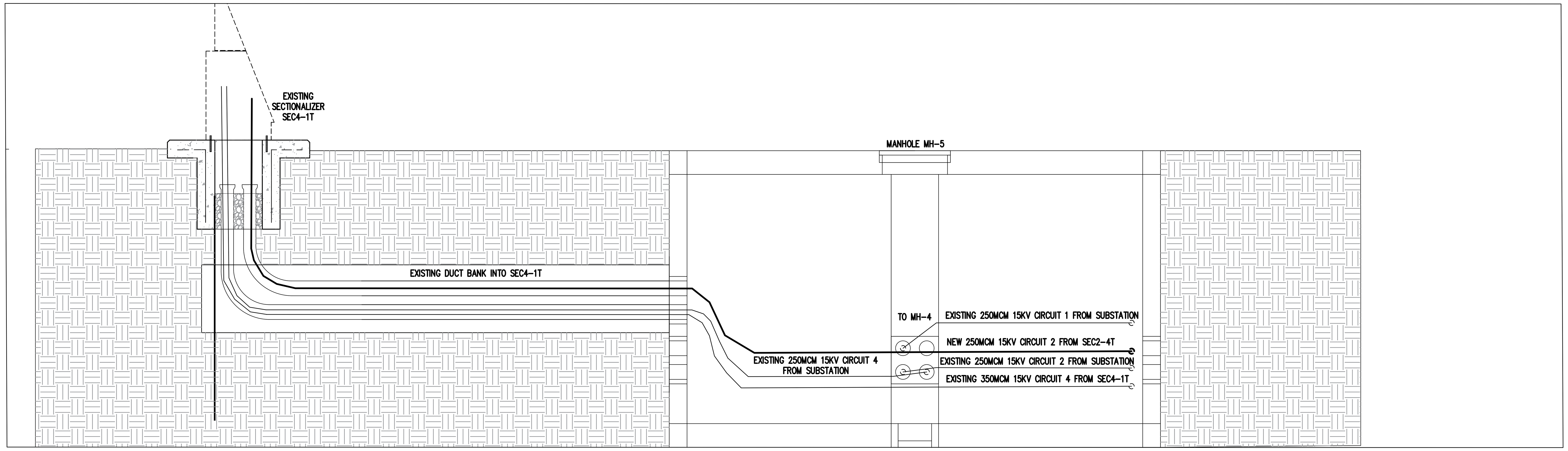
DRAWN BY: RLS
CHECKED BY: RLS
PROJECT NO.: XTLF 25-1025
DATE: 04 APR 2025
DESIGN STAGE: FOR CONSTRUCTION

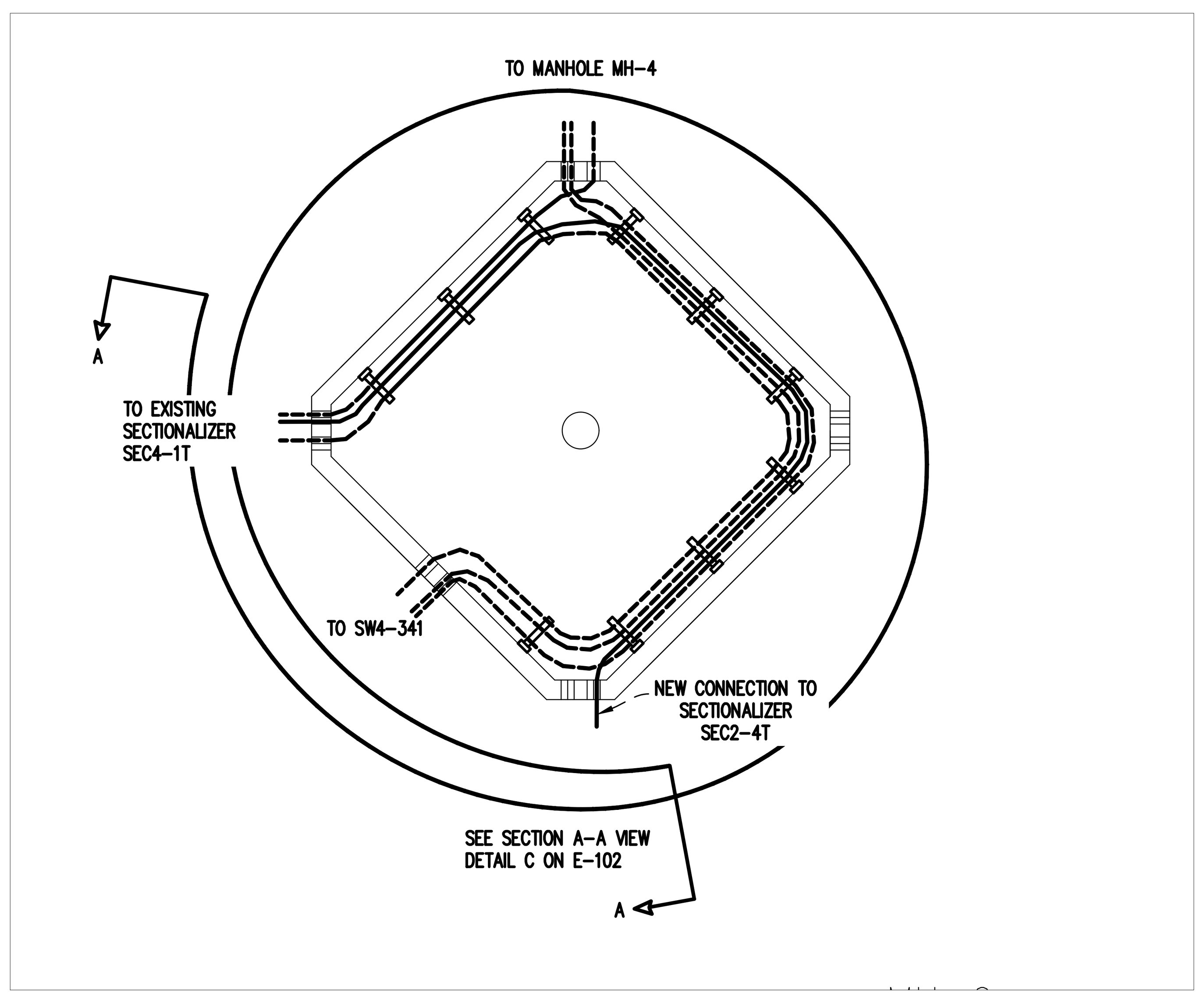
ELECTRICAL DISTRIBUTION RESILIENCE CIRCUIT 2 LOOP FEED

SWITCH DETAILS

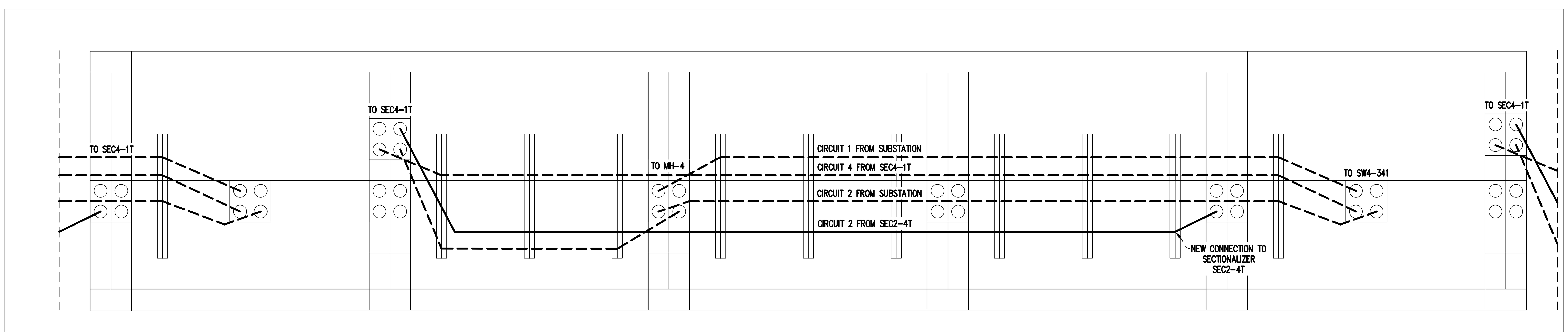
E501



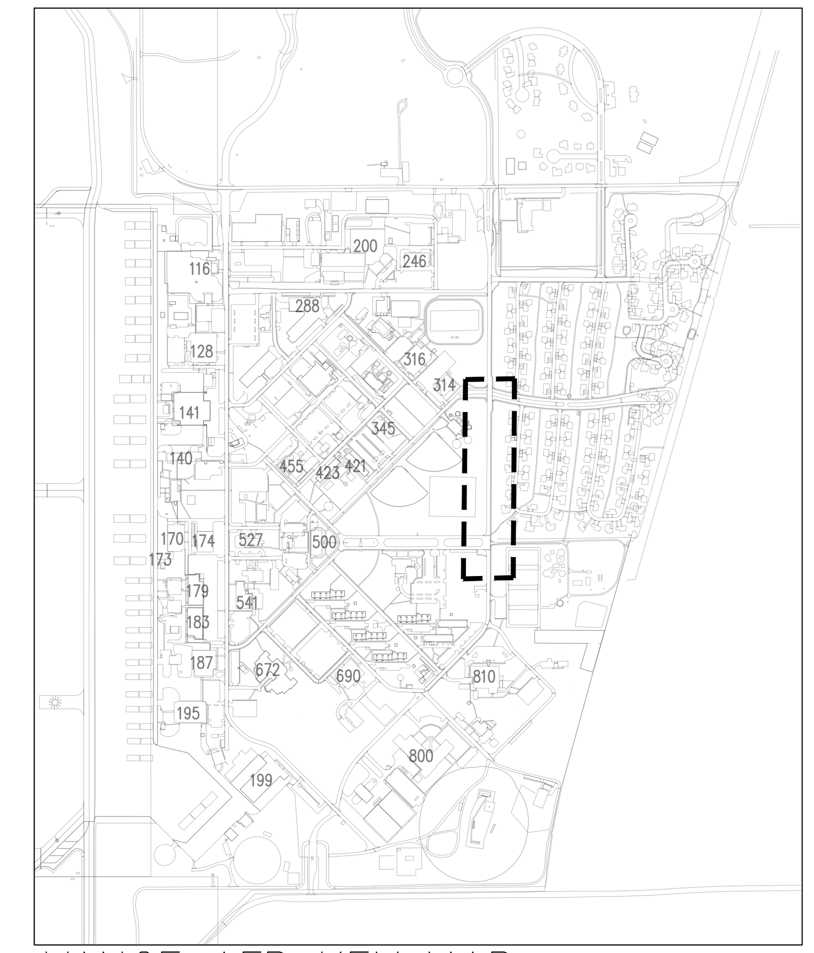
A E502 MANHOLE MH-5 LAYOUT
NOT TO SCALE



B E502 MANHOLE MH-5 PLAN VIEW
NOT TO SCALE



C E502 MANHOLE WALL SECTION A-A VIEW
NOT TO SCALE



VANCE AFB KEY MAP
NOT TO SCALE

GENERAL NOTES

1. ALL WIRE SHOWN ON THIS SHEET BETWEEN SWITCHES AND/ OR 600A SECTIONALIZERS SHALL BE 250 MCM CU AND SHALL BE IN 5-IN SCHEDULE 40 PVC CONDUIT FOR OPEN TRENCH INSTALLATION OR 5-IN SDR9 HDPE FOR HORIZONTAL DIRECTIONAL DRILL INSTALLATION..
2. ALL CONDUIT INSTALLED IN THIS PROJECT SHALL HAVE A MINIMUM OF AT LEAST ONE SPARE EMPTY CONDUIT INSTALLED OF THE SAME SIZE, CONNECTING TO THE SAME EQUIPMENT. PULL STRING SHALL BE INCLUDED IN ALL SPARE DUCTS.
3. ALL CONDUCTORS IN MANHOLE MH-5 TO BE DE-ENERGIZED AND STATIC CHARGE TO BE BLED OFF PRIOR TO COMMENCING WORK IN MANHOLE MH-5.

UNITED STATES AIR FORCE
AIR EDUCATION
TRAINING COMMAND
VANCE AIR FORCE BASE
ENID, OKLAHOMA

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JUNE 30, 2026

Revision Number	Description	Date

LICENSED PROFESSIONAL ENGINEER
RYAN SUDLOW
31220
OKLAHOMA

DRAWN BY: RLS
CHECKED BY: RLS
PROJECT NO.: XTLF 25-1025
DATE: 04 APR 2025
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ELECTRICAL DISTRIBUTION RESILIENCE CIRCUIT 2 LOOP FEED

MAN HOLE MH-5 DETAILS

E102P